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OFFICE OF CHEMICAL SAFETY
AND POLLUTION PREVENTION

OFFICE OF PESTICIDE PROGRAMS
REGISTRATION DIVISION (7505P)

May 10, 2017

MEMORANDUM

SUBJECT: Naphthalenesulfonate Formaldehyde Condensates Potassium Salts; Human Health Risk Assessment and Ecological Effects Assessment to Support Proposed Exemption from the Requirement of a Tolerance When Used as an Inert Ingredient in Pesticide Formulations.

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Regulatory Action: Inert Tolerance
Exemption; 40 CFR 180.910

All CAS Reg. No.: 67828-14-2

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EXECUTIVE SUMMARY

Monsanto (1300 I (Eye) Street, NW Suite 450 East, Washington, DC 20005) is requesting that the existing 40 CFR 180.910 exemption from the requirement of a tolerance for residues of sodium and ammonium naphthalenesulfonate formaldehyde condensates (SANFCs) when used as inert ingredients (surfactants and related adjuvants of surfactants) in pesticide formulations applied to growing crops and raw agricultural commodities after harvest be amended to include naphthalenesulfonate formaldehyde condensates potassium salt (CAS Reg. No. 67828-14-2) (also referred to as potassium salt of mono-, di-, and trimethylnaphthalenesulfonic acid formaldehyde condensates). EPA published the notice of filing (NOF) for this petition in the Federal Register on October 18, 2016 (81 FR 71668).

The Agency has reviewed Monsanto's claim that the addition of the potassium salt of mono-, di-, and trimethylnaphthalenesulfonic acids and naphthalenesulfonic acids formaldehyde condensates (CAS Reg. No. 67828-14-2) is acceptable under the current tolerance exemption descriptor for Mono-, di-, and trimethylnaphthalenesulfonic acids and naphthalenesulfonic acids formaldehyde condensates, ammonium and sodium salts. The Chemistry, Inerts, and Toxicology Assessment Branch (CITAB) has confirmed that the requested CAS Reg. No. is acceptable for consideration under the currently approved tolerance exemption descriptor.

As a class, the ammonium, potassium and sodium naphthalenesulfonate formaldehyde condensates are not expected to be acutely toxic, mutagenic, or genotoxic, nor a skin irritant.

In a 28-42 day OECD Test Guideline 422 combined repeated dose toxicity study with the reproduction/developmental toxicity screening with naphthalenesulfonic acid, sodium salt polymer with formaldehyde (CAS 9084-06-4), there was no evidence of increased susceptibility (MRID 47643301). Parental toxicity manifested as decrements in body-weight gain in both sexes at the limit dose (1000 mg/kg/day). No developmental or reproductive effects were observed at doses of 100, 300, and 1000 mg/kg/day.

The International Uniform Chemical Information Database (IUCLID) for Naphthalenesulfonic acid, polymer with formaldehyde, sodium salt (CAS Reg. No. 9084-06-4) reported that an Ames test was negative, confirming the absence of mutagenic potential for this class of compounds.

There is no evidence that the SANFCs are carcinogenic. The Agency used a qualitative structure activity relationship (SAR) database, DEREK Version 11, to determine if there were structural alerts. No structural alerts were identified and there are no target organs of toxicity. In addition, there was little concern about any of the postulated metabolites having greater toxicity than the parent compounds.

There was no evidence of potential immunotoxicity in the adult animal in the OECD 422 study at the limit dose of 1000 mg/kg/day.

There was no evidence of potential neurotoxicity in the adult animal in the OECD 422 study at the limit dose of 1000 mg/kg/day.

The higher molecular weight polymeric SANFC surfactants (MW>1000) are not expected to be readily absorbed or metabolized, and should thus be rapidly excreted (likely in the feces) unchanged. By analogy to other classes of similar surfactants, the SANFC naphthalene ring sodium sulfonate group could provide a handle for conjugation and excretion. Additionally, lower molecular microsome cytochrome P-450 oxygenases may hydroxylate the naphthalene ring and/or methylene bridge to produce alternative metabolites that should also be readily conjugated and excreted.

There are no dermal absorption data on the SANFCs; however, given the molecular weights and solubility characteristics of these chemicals, dermal absorption is expected to be low.

There was no significant hazard identified in the OECD 422 study at the limit dose of 1000 mg/kg/day to either parental animals or their offspring. Thus, due to their low potential hazard and the lack of a hazard endpoint, it was determined that a quantitative risk assessment using safety factors applied to a point of departure protective of an identified hazard endpoint is not appropriate for the SANFCs. There was no evidence of neurotoxicity or increased susceptibility to the offspring of rats following prenatal or postnatal exposure in the OECD 422 study. For the same reasons that a quantitative risk assessment based on a safety factor approach is not appropriate for the SANFCs, an FQPA SF is not needed to protect the safety of infants and children. Additionally, residential and occupational exposure assessments were not conducted.

As a class, the lower molecular weight/bioavailable non-polymeric ammonium, potassium and sodium naphthalenesulfonate formaldehyde condensate surfactants are expected to be nonvolatile, have high melting points, not lipophilic with increasing log K_{ow} with increasing methylation, bind tightly to soil, undergo oxidative degradation within 8 hours and biodegrade with affinity for water and soil compartments.

Potential areas of environmental justice concerns, to the extent possible, were considered in this human health risk assessment, in accordance with U.S. Executive Order 12898, "Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations," <http://www.eh.doe.gov/oepa/guidance/justice/eo12898.pdf>.

Taking into consideration all available information on SANFCs in regards to naphthalenesulfonate formaldehyde condensate potassium salts (CAS Reg. No. 67828-14-2), EPA concludes that there is a reasonable certainty that no harm will result to the general population when naphthalenesulfonate formaldehyde condensate potassium salts is used as an inert ingredient. The amendment of the exemption from tolerance for residues of sodium and ammonium naphthalenesulfonate formaldehyde condensates (SANFCs) to include naphthalenesulfonate formaldehyde condensate potassium salts when used as an inert ingredient (surfactant and related adjuvant of surfactants) applied pre- and post-harvest under 40 CFR 180.910 in the pesticide formulation can be considered safe under section 408 of the FFDCa.

I. BACKGROUND

EPA received petition (IN-10965) from Monsanto (1300 I (Eye) Street, NW Suite 450 East, Washington, DC 20005) requesting that the existing 40 CFR 180.910 exemption from the requirement of a tolerance for residues of sodium and ammonium naphthalenesulfonate formaldehyde condensates (SANFCs) when used as inert ingredients (surfactants and related adjuvants of surfactants) in pesticide formulations applied to growing crops and raw agricultural commodities after harvest be amended to include naphthalenesulfonate formaldehyde condensates potassium salt (CAS Reg. No. 67828-14-2) (also referred to as potassium salt of mono-, di-, and trimethylnaphthalenesulfonic acid formaldehyde condensates). EPA published the notice of filing (NOF) for this petition in the Federal Register on October 18, 2016 (81 FR 71668).

On October 7, 2009, EPA published a Final Rule in the Federal Register (74 FR 51470) granting an exemption from the requirement of a tolerance for residues of the sodium and ammonium naphthalenesulfonate formaldehyde condensates (SANFCs) when used as inert ingredients (surfactants and related adjuvants of surfactants) in pesticide formulations applied to growing crops and raw agricultural commodities after harvest (PP 9E7572 and 9E7562). Monsanto is petitioning the Agency to amend the exemption from tolerance for SANFCs to include the naphthalenesulfonate formaldehyde condensates potassium salt (also referred to as potassium salt of mono-, di-, and trimethylnaphthalenesulfonic acids and naphthalenesulfonic acids formaldehyde condensates) (CAS Reg. No. 67828-14-2).

The Agency has reviewed Monsanto's claim that the addition of the potassium salt of mono-, di-, and trimethylnaphthalenesulfonic acids and naphthalenesulfonic acids formaldehyde condensates (CAS Reg. No. 67828-14-2) is acceptable under the current tolerance exemption descriptor for Mono-, di-, and trimethylnaphthalenesulfonic acids and naphthalenesulfonic acids formaldehyde condensates, ammonium and sodium salts. The Chemistry, Inerts, and Toxicology Assessment Branch (CITAB) has confirmed that the requested CAS Reg. No. is acceptable for consideration under the currently approved tolerance exemption descriptor.

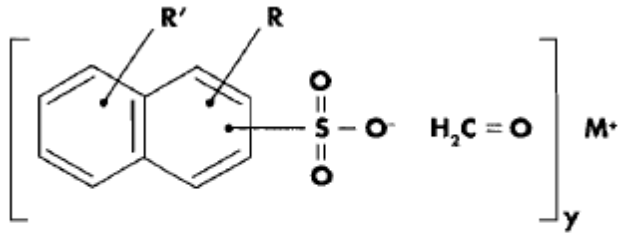
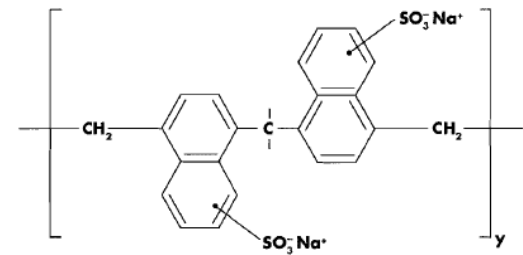
II. INERT INGREDIENT PROFILE

A. Summary of Uses

SANFC inert ingredients are used as dispersants, defoamers and emulsifiers in pesticide formulations. These surfactants have a wide range of industrial uses such as dispersant for pigments, latex and natural fibers, defoamer in food contact packaging, emulsifier in personal care products, and in cellophane for food packaging.

Physical Chemical Properties

SANFCs are anionic surfactants that may be derived from naphthalene or methyl naphthalenesulfonic acids. The surfactants included in this group are complex mixtures and some constituents of these mixtures are likely to be very large molecules (Table 1 and 2).

Table 1. Structures of Naphthalenesulfonate Formaldehyde Condensates		
Parameter	Values	Reference
General Chemical Structure ¹	 $M^+ = \text{Na}^+, \text{NH}_4^+$ $R, R' = \text{H or CH}_3 \text{ and combinations of H and CH}_3$ $y = \gg 1$	EPA 2009
Structure of Main Component of Test Material in the Toxicity Testing	 $y = \gg 1$	EPA 2009

¹ Figures excerpted from petition, “Petition Proposing An Exemption From The Requirement Of A Tolerance For Residues Of Joint Inerts Task Force Cluster 11 Sodium And Ammonium Naphthalenesulfonate Formaldehyde Condensates In Or On Raw Agricultural Products And Food Product”, pages 6-7, dated January 9, 2009

Table 2. Listing of Chemicals included in the SANFC Cluster	
CAS Number	Descriptor
9008-63-3	Naphthalenesulfonic acid, sodium salt polymer with formaldehyde
9069-80-1	Naphthalenesulfonic acid, ammonium salt polymer with formaldehyde
9084-06-4	Naphthalenesulfonic acid, polymer with formaldehyde, sodium salt
36290-04-7	2-Naphthalenesulfonic acid, polymer with formaldehyde, sodium salt
91078-68-1	Naphthalenesulfonic acids, reaction products with formaldehyde, sodium salts
141959-43-5	Naphthalenesulfonic acid, methyl-sodium salt with formaldehyde
68425-94-5	Residues, petroleum, catalytic reformer fractionator, sulfonated, polymers with formaldehyde, sodium salts

Table excerpted from Human Health Risk Assessment, “Sodium and Ammonium Naphthalenesulfonate Formaldehyde Condensates (SANFCs - JITF CST 11 Inert Ingredients). Human Health Risk Assessment to Support Proposed Exemption from the Requirement of a Tolerance When Used as Inert Ingredients in Pesticide Formulations.”, page 5, dated May 28, 2009

Physicochemical data were estimated for the SANFC inert ingredients using publicly available Estimation Program Interface suite (EPI Suite™) modeling (<http://www.epa.gov/opptintr/exposure/pubs/episuite.htm>) and are provided in Table 3. The basis for the estimated EPI Suite™ modeling physicochemical calculations and model capabilities are described in the EPA document titled The Use of Structure-Activity Relationships (SAR) in the High Production Volume Chemicals Challenge Program (EPA 1999). The EPA EPI Suite™ 3.2 model calculations for water solubility are based on Kow (WSKOWWIN) and molecular fragments (WATERNT). EPA has cautioned in the publication OPPTS Pollution Prevention P2

Network (EPA, 2005), that EPI Suite™ 3.2 water solubility calculations are not reliable for dispersible compounds such as surfactants.

Table 3. Physicochemical Properties of Representative SSNA Compounds ¹						
CAS No.	Modeled As	Log K_{ow}	Boiling Point (°C)	Melting Point (°C)	Vapor Pressure (mm Hg)	Water Solubility (mg/L) ²
91078-68-1	Condensate Dimer	-0.32	854.3	349.8	1.01 x 10 ⁻¹⁸	155.3 3.99x10 ⁻⁵
Methyl example?	Methylnaphthalene sulfonate	0.78	877.5	349.8	1.53 x 10 ⁻¹⁵	11.95 3.76x10 ⁻⁶

¹ Table values taken from submission entitled “Petition Proposing An Exemption From The Requirement Of A Tolerance For Residues Of Joint Inerts Task Force Cluster 11 Sodium And Ammonium Naphthalenesulfonate Formaldehyde Condensates In Or On Raw Agricultural Products And Food Product”, 1/29/2009, Table 2, p. 13.

² The EPI Suite™ model calculates water solubility based on the octanol-water partition coefficient (Kow) (WSKOWWIN) and molecular fragments (WATERNT). The first value reflects the WSKOWWIN calculation and the second value reflects the WATERNT calculation.

As a class, the lower molecular weight/bioavailable non-polymeric ammonium, potassium and sodium naphthalenesulfonate formaldehyde condensate surfactants have the following physicochemical profile:

- Nonvolatile
- High melting point
- Not lipophilic, but with increasing log Kow with increasing methylation, going to 0.78 from -0.32 for CAS Reg. No. 91078-68-1 (no methylation)
- Binds tightly to soil
- Oxidative degradation within 8 hours
- Biodegradable with affinity for water and soil compartments

III. HUMAN HEALTH ASSESSMENT

A. Summary of Toxicity Data

The existing toxicology database for the SANFC consists of one OECD 422 combined repeated dose toxicity study with the reproduction/developmental toxicity screening in rats on the representative SANFC, and several studies from the scientific literature on acute toxicity and mutagenicity.

The OECD Test Guideline 422 study provides information on possible health hazards likely to arise from repeated exposure over a relatively limited period of time. The study is comprised of basic repeated dose toxicity study, a reproduction/developmental toxicity screening test, and places emphasis on neurological effects as a specific endpoint. Therefore, the study provides information on general toxicity as well as possible developmental, reproductive, and neurological effects. It also provides a basic indications of immunological effects.

Acute Toxicity

CAS Reg. No. 91078-68-1 is nontoxic, with a rat oral LD₅₀ of 6,168 mg/kg and an LD₅₀ of >5,000 mg/kg in mice, when tested as liquid commercial formulations in water with a solid content of around 50%, and acute inhalation LC₅₀ of around 50 mg/L.

For CAS Reg. No. 9084-06-4, The Handbook of Industrial Surfactants (Volume 1 and 2) reports that the rat oral LD₅₀ is 3,800 mg/kg. Alternate routes of exposure to rats resulted in an LD₅₀ of 460 mg/kg (IP) and 435 mg/kg (IV). Subacute mouse LD₅₀ is reported to be 1,275 mg/kg.

In addition, CAS Reg. No. 9084-06-4 did not reveal any irritating properties when applied to the skin of rabbits in several studies, and was at worst slightly irritating to the rabbit eye.

Repeat Dose Toxicity –

In a repeated 28-42 day OECD Test Guideline 422 combined repeated dose toxicity study with the reproduction/developmental toxicity screening in rats with the representative test compound, naphthalenesulfonic acid, sodium salt polymer with formaldehyde (CAS 9084-06-4), there was no evidence of increased susceptibility (MRID 47643301). Parental toxicity manifested as decrements in body-weight gain in both sexes at the limit dose (1000 mg/kg/day). No developmental or reproductive effects were observed at doses of 100, 300, and 1000 mg/kg/day.

There was no evidence of potential neurotoxicity or immunotoxicity in the adult animal in the OECD 422 study at the limit dose of 1000 mg/kg/day. There are no chronic data available on the SANFC surfactants; however, no structural alerts for cancer were identified in a qualitative structure activity relationship (SAR) database, DEREK Version 11.

Mutagenicity and Genotoxicity

The IUCLID for CAS Reg. No. 9084-06-4 reported that an Ames test was negative, indicating an absence of mutagenic potential for this class of compounds.

Reproductive and Developmental Toxicity

In the study, naphthalenesulfonic acid, polymer with formaldehyde sodium salt (CAS Reg. No. 9084-06-4) Combined Repeated Dose Toxicity Study with the Reproduction/ Developmental Toxicity Screening Test in the Han Wistar Rat. Harlan study B74147 (MRID 47643301). The overall NOAEL (No Observed Adverse Effect Level) was established at 1000 mg/kg body weight/day. The only effects observed at this dosage level were transient effects on food consumption and body weight at the beginning of treatment that were not considered adverse because the animals recovered immediately thereafter, and there was no impact on the terminal mean body weight of both males and females treated at the limit dose of 1000 mg/kg/day. The NOAEL for reproduction/ developmental toxicity was considered to be 1000 mg/kg/day, with the single occurrence of total litter loss at this dosage level considered to be within historical control range.

Carcinogenicity

There is no evidence that the SANFCs are carcinogenic. The Agency used a qualitative structure activity relationship (SAR) database, DEREK Version 11, to determine if there were structural alerts. No structural alerts were identified. There were no target organs of toxicity. In addition, there was little concern about any of the postulated metabolites having greater toxicity than the parent compounds.

Immunotoxicity

There was no evidence of potential immunotoxicity in the adult animal in the OECD 422 study at the limit dose of 1000 mg/kg/day.

Neurotoxicity

There was no evidence of potential neurotoxicity in the adult animal in the OECD 422 study at the limit dose of 1000 mg/kg/day.

Metabolism and Pharmacokinetics

As stated in the petition from JITF Cluster Support Team 11, the higher molecular weight polymeric SANFC surfactants (MW>1000) are not expected to be readily absorbed or metabolized, and should thus be rapidly excreted (likely in the feces) unchanged. By analogy to other classes of similar surfactants, the SANFC naphthalene ring sodium sulfonate group could provide a handle for conjugation and excretion. Additionally, lower molecular microsome cytochrome P-450 oxygenases may hydroxylate the naphthalene ring and/or methylene bridge to produce alternative metabolites that should also be readily conjugated and excreted.

Dermal Absorption

There are no dermal absorption data on the SANFCs; however, given the molecular weights and solubility characteristics of these polymers, dermal absorption is expected to be low.

B. Toxicity Endpoint Selection and FQPA Considerations

There was no significant hazard identified in the OECD 422 study at the limit dose of 1000 mg/kg/day to either parental animals or their offspring. Thus, due to their low potential hazard and the lack of a hazard endpoint, it was determined that a quantitative risk assessment using safety factors applied to a point of departure protective of an identified hazard endpoint is not appropriate for the SANFCs. There was no evidence of neurotoxicity or increased susceptibility to the offspring of rats following prenatal or postnatal exposure in the OECD 422 study. For the same reasons that a quantitative risk assessment based on a safety factor approach is not appropriate for the SANFCs, an FQPA SF is not needed to protect the safety of infants and children. Based on the close structural similarity of sodium and ammonium naphthalenesulfonate formaldehyde condensates to naphthalenesulfonate formaldehyde condensates potassium salt, differing

only in the respective compounds' counterion (i.e., potassium versus sodium or ammonium), the toxicity profile and toxicity endpoints would be equivalent.

IV. EXPOSURE ASSESSMENT

A. Dietary Exposure:

i. Acute exposure. No adverse effects attributable to a single exposure of SANFCs was seen in the toxicity databases. Therefore, acute dietary risk assessment for SANFCs is not required.

ii. Chronic exposure. No adverse effects attributable to a repeat dose exposure of SANFCs was seen in the toxicity databases. Therefore, chronic dietary risk assessment for SANFCs is not required.

iii. Cancer exposure. There is no carcinogenicity concern from dietary exposures to SANFCs.

B. Residential (Non-Occupational) Exposure:

These inert ingredients may be used in personal care products. However the lack of concern for hazard posed by the SANFC inert ingredients, the Agency concludes that there are no residential (non-dietary/non-occupational) risks of concern for these inert ingredients.

V. AGGREGATE RISK ASSESSMENT

In examining aggregate exposure, FFDCa section 408 directs EPA to consider available information concerning exposures from the pesticide residue in food and all other non-occupational exposures, including drinking water from ground water or surface water and exposure through pesticide use in gardens, lawns, or buildings (residential and other indoor uses). FFDCa section 408 provides that EPA shall also apply an additional tenfold margin of safety for infants and children in the case of threshold effects in calculating a dose level that to account for pre- and post-natal toxicity and the completeness of the database unless EPA determines that a different margin of safety will be safe for infants and children. Margins of safety are incorporated into EPA risk assessments either directly through the use of margin of exposure analysis or through using uncertainty (safety) in calculating a dose level that poses no appreciable risk to humans.

i. Acute Dietary Risk.

An acute aggregate risk assessment takes into account exposure estimates from acute dietary consumption of food and drinking water using the exposure assumptions discussed in Section IV for acute exposure. However there was no hazard attributable to

a single exposure seen in the toxicity database for SANFCs; therefore there is no concern for an acute dietary risk and subsequently an acute aggregate risk assessment was not conducted.

ii. Chronic Dietary Risk.

A chronic aggregate risk assessment takes into account exposure estimates from chronic dietary consumption of food and drinking water using the exposure assumptions discussed in Section IV from chronic exposure. However there was no hazard attributable to a repeat dose exposure seen in the toxicity database for SANFCs; therefore there is no concern for a chronic dietary risk and as a result a chronic aggregate risk assessment was not conducted.

iii. Short-term Aggregate Risk.

Short-term aggregate exposure takes into account short-term residential exposure plus chronic exposure to food and water (considered to be a background exposure level). SANFCs may be used as inert ingredients in pesticide products that could result in short-term residential exposure. However, chronic dietary risk was not assessed due to the absence of any hazards identified in the toxicity database. Therefore short-term aggregate risk for SANFCs is considered to be not of concern.

iv. Intermediate-Term Aggregate Risk.

Intermediate -term aggregate exposure takes into account intermediate-term residential exposure plus chronic exposure to food and water (considered to be a background exposure level). SANFCs may be used as inert ingredients in pesticide products that could result in intermediate-term residential exposure. However, chronic dietary risk was not assessed due to the absence of any hazards identified in the toxicity database. Therefore intermediate-term aggregate risk for SANFCs is considered to be not of concern.

v. Aggregate Cancer Risk for U.S. Population.

The Agency has concluded that use of SANFCs is not likely to result in any exposures that may pose cancer concerns; therefore, an aggregate cancer risk assessment was not conducted.

VI. OCCUPATIONAL EXPOSURE/RISK PATHWAY

The representative occupational scenarios selected by the Agency for assessment are evaluated based on likely maximum application rates for products which may contain the inert ingredient for the short-term exposure assessment, and average application rates for products likely to contain the inert ingredient for the intermediate- and long-term exposure durations. Active ingredient application rates are corrected for the maximum amount of inert ingredient likely to be in the final formulations to determine exposure and risk from exposure to the inert ingredient by fungicide/insecticide or herbicide.

The Agency traditionally considers a level of concern (LOC) for these risk assessments to

be an MOE of 100 based on the standard 10X inter and 10X intra species extrapolation safety factors.

A. Occupational Handler Risk

Exposure Scenarios

Dermal and inhalation exposure are estimated using the Pesticide Handlers Exposure Database (PHED) and Outdoor Residential Exposure Task Force (ORETF) data. The analytical techniques are those developed by the Science Policy Council on Exposure (ExpoSAC), Health Effects Division, Office of Pesticide Programs, US Environmental Protection Agency. The quantitative exposure/risk assessment developed for occupational handlers to support the requested exemption for the inert ingredient based on scenarios to represent the highest potential exposure.

Since no endpoints of concern were identified in the toxicity database for the SANFC inert ingredients, a quantitative occupational handler risk assessment was not performed. Due to the low hazard associated with these inert ingredients, worker risks are not of concern.

B. Occupational Post-Application Risk

Exposure Scenarios

RD uses the term post-application to describe exposures that occur when individuals are present in an environment that has been previously treated with a pesticide (also referred to as re-entry exposure). Such exposures may occur when workers enter previously treated areas to perform job functions, including activities related to crop production, such as scouting for pests or harvesting. Post-application exposure levels vary over time and depend on such things as the type of activity, the nature of the crop or target that was treated, the type of pesticide application, and the chemical's degradation properties. In addition, the timing of pesticide applications, relative to harvest activities, can greatly reduce the potential for post-application exposure. Inhalation exposures are also not typically calculated for occupational post-application scenarios because inhalation exposures generally account for a negligible percentage of the overall body burden for most pesticide chemicals.

Since no endpoints of concern were identified in the toxicity database for the SANFC inert ingredients, a quantitative occupational post-application risk assessment was not performed. Due to the low hazard associated with these inert ingredients, worker risks are not of concern.

VII. CUMULATIVE EXPOSURE

Cumulative effects from substances with a common mechanism of toxicity. Section 408(b)(2)(D)(v) of FFDCFA requires that, when considering whether to establish, modify, or

revoke a tolerance, the Agency consider “available information” concerning the cumulative effects of a particular pesticide's residues and “other substances that have a common mechanism of toxicity.”

EPA has not found SANFCs to share a common mechanism of toxicity with any other substances, and SANFCs do not appear to produce a toxic metabolite produced by other substances. For the purposes of this tolerance action, therefore, EPA has assumed that SANFCs do not have a common mechanism of toxicity with other substances. For information regarding EPA's efforts to determine which chemicals have a common mechanism of toxicity and to evaluate the cumulative effects of such chemicals, see EPA's website at <http://www.epa.gov/pesticides/cumulative>.

VIII. ENVIRONMENTAL JUSTICE STATEMENT

Potential areas of environmental justice concerns, to the extent possible, were considered in this human health risk assessment, in accordance with U.S. Executive Order 12898, "Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations," <http://www.eh.doe.gov/oepa/guidance/justice/eo12898.pdf>.

As a part of every pesticide risk assessment, Office of Pesticide Program (OPP) considers a large variety of consumer subgroups according to well-established procedures. In line with OPP policy, Registration Division (RD) estimates risks to population subgroups from pesticide exposures that are based on patterns of that subgroup's food and water consumption, and activities in and around the home that involve pesticide use in a residential setting. Extensive data on food consumption patterns are compiled by the USDA under CSFII and are used in pesticide risk assessments for all registered food uses of a pesticide. These data are analyzed and categorized by subgroups based on age, season of the year, ethnic group, and region of the country. Additionally, OPP is able to assess dietary exposure to smaller, specialized subgroups and exposure assessments are performed when conditions or circumstances warrant. Whenever appropriate, non-dietary exposures based on home use of pesticide products and associated risks for adult applicators and for toddlers, youths, and adults entering or playing on treated areas post-application are evaluated. Further considerations are currently in development as OPP has committed resources and expertise to the development of specialized software and models that consider exposure to bystanders and farm workers as well as lifestyle and traditional dietary patterns among specific subgroups.

IX. ENVIRONMENTAL FATE CONSIDERATIONS

As a class, the lower molecular weight/bioavailable non-polymeric ammonium, potassium and sodium naphthalenesulfonate formaldehyde condensate surfactants have the following physicochemical profile:

- Nonvolatile
- High melting point

- Not lipophilic, but with increasing log Kow with increasing methylation, going to 0.78 from -0.32 for CAS Reg. No. 91078-68-1 (no methylation)
- Binds tightly to soil
- Oxidative degradation within 8 hours
- Biodegradable with affinity for water and soil compartments

X. ECOTOXICITY

No data submitted.

XI. RISK CHARACTERIZATION

There was no significant hazard identified in any of the studies up to the limit dose of 1000 mg/kg/day to either parental animals or their offspring. Thus, due to their low potential hazard and the lack of a hazard endpoint, it was determined that a quantitative risk assessment using safety factors applied to a point of departure protective of an identified hazard endpoint is not appropriate for the SANFCs. There was no evidence of neurotoxicity or increased susceptibility to the offspring of rats following prenatal or postnatal exposure in the OECD 422 study. For the same reasons that a quantitative risk assessment based on a safety factor approach is not appropriate for the SANFCs, an FQPA SF is not needed to protect the safety of infants and children. Additionally residential and occupational exposure assessments were not conducted.

As a class, the lower molecular weight/bioavailable non-polymeric ammonium, potassium and sodium naphthalenesulfonate formaldehyde condensate surfactants are expected to be nonvolatile, have high melting points, not lipophilic with increasing log Kow with increasing methylation, bind tightly to soil, undergo oxidative degradation within 8 hours and biodegrade with affinity for water and soil compartments.

Taking into consideration all available information on SANFCs in regards to naphthalenesulfonate formaldehyde condensate potassium salts (CAS Reg. No. 67828-14-2), EPA concludes that there is a reasonable certainty that no harm will result to the general population when naphthalenesulfonate formaldehyde condensate potassium salts is used as an inert ingredient. The establishment of an exemption from tolerance for food use for residues of naphthalenesulfonate formaldehyde condensate potassium salts as an inert ingredient surfactant and related adjuvant of surfactants applied pre- and post-harvest under 40 CFR 180.910 in the pesticide formulation can be considered safe under section 408 of the FFDCA.

REFERENCES

Monsanto Company (2016) Petition to Amend the Exemption From the Requirement of a Tolerance for Pre- and Post-Harvest Residues of Sodium and Ammonium Naphthalenesulfonate Formaldehyde Condensates In or On Raw Agricultural Commodities and Food Products to Include: Potassium Naphthalenesulfonate Formaldehyde Condensates.

U.S. EPA (2009) Mono-, Di-, Trimethylnaphthalenesulfonic Acids and Naphthalenesulfonic Acids Formaldehyde Condensates, Ammonium and Sodium Salts: Review of Toxicological Studies in Support of an Exemption from the Requirement of a Tolerance (40 CFR 180.920 and 40 CFR 180.910) When Used as Inert Ingredients in Pesticide Formulations.

U.S. EPA (2009) Sodium and Ammonium Naphthalenesulfonate Formaldehyde Condensates (SANFCs – JITF CST 11 Inert Ingredients). Human Health Risk Assessment to Support Proposed Exemption from the Requirement of a Tolerance When Used as Inert Ingredients in Pesticide Formulations.